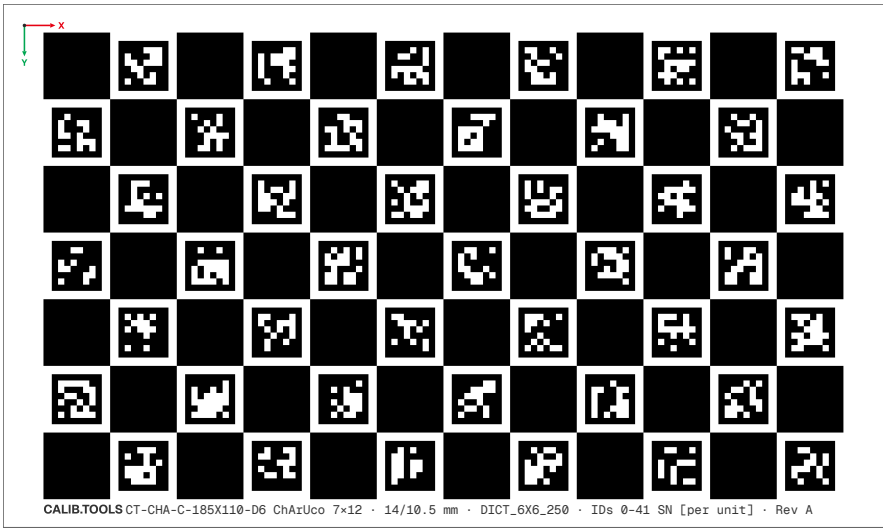
DATASHEET

# ChArUco Calibration Target, Alumina Ceramic

A rigid technical-ceramic ChArUco target for precise camera calibration where dimensional stability, sharp features and durable handling matter.




Rendering · 185 × 110 mm variant

COORDINATE SYSTEM




(0, 0) · pattern outer corner

X right · Y down
OpenCV pattern frame

## 1 Ordering

PRINT VERSION · prices online

| Part number | Size W × H (mm) | Squares (r × c) | Square / marker (mm) | Dictionary options | Corners | Markers | Weight (kg) |
|---|---|---|---|---|---|---|---|
| CT-CHA-C-110085-6X6 | 110 × 85 | 7 × 10 | 10 / 7.5 | DICT_6X6_250, DICT_6X6_1000 | 54 | 35 | 0.10 |
| CT-CHA-C-150100-6X6 | 150 × 100 | 7 × 11 | 12.5 / 9.5 | DICT_6X6_250, DICT_6X6_1000 | 60 | 38 | 0.17 |
| CT-CHA-C-185110-6X6 | 185 × 110 | 7 × 12 | 14 / 10.5 | DICT_6X6_250, DICT_6X6_1000 | 66 | 42 | 0.22 |
| CT-CHA-C-230140-6X6 | 230 × 140 | 8 × 14 | 15 / 11 | DICT_6X6_250, DICT_6X6_1000 | 91 | 56 | 0.35 |

All variants: white margin · alumina ceramic, 3 mm. See ordering table for per-size marker dimensions.

Prices: calib.tools/targets

## 2 Key specifications

ACCURACY
±2 µm
feature position · 20 °C

SUBSTRATE
Technical ceramic
alumina, 3 mm

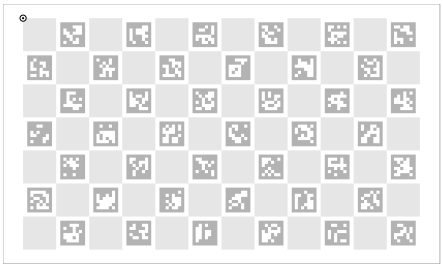
SIZES
110 × 85 to 230 × 140 mm

FLATNESS
≤ 20 µm
over pattern area

Unless stated otherwise, dimensions are nominal at 20 °C. Specifications apply to the product family.

## 1 Specifications · geometry

| FEATURE | SPECIFICATION | CONDITION / NOTE |
|---|---|---|
| Feature position accuracy | ±2 µm | At 20 °C; ChArUco corner positions. |
| Edge definition | Sharp black / white transition | ≤ 1 µm lithographic edge. |
| Orthogonality | 90° ± 0.002° | Pattern axes. |
| Start ID / placement | 0 · row-major | Black square top-left; marker IDs advance through white squares. |
| OpenCV pattern convention | OpenCV ≥ 4.6 layout | Odd-row board: no legacy-pattern issue with OpenCV ≥ 4.6 or older versions. |



Geometry reference · pattern envelope shown.

PATTERN CONVENTION

| | |
|---|---|
| Reference pattern | 7 × 12 squares · 14 / 10.5 mm. |
| Marker IDs | 0–41, row-major through white squares. |
| Coordinate frame | Origin at the top-left outer pattern corner; X right, Y down. |
| Version control | Odd-row board: no legacy-pattern issue with OpenCV ≥ 4.6 or older versions. |

## 2 Specifications · physical

| FEATURE | SPECIFICATION |
|---|---|
| Substrate | Alumina ceramic ($Al_2O_3$ ≥ 96 %), white, 3 mm. |
| Pattern process | Lithographed chrome; blue-black, low gloss. |
| Flatness | ≤ 20 µm over the pattern area. |
| Surface | Ra ≤ 0.4 µm; lapped, non-glare. |
| Edges | 0.5 mm chamfer, ground. |
| Weight | ≈ 11 kg/m²; 150 × 100: 0.17 kg. |
| Mounting | No holes. Use CT-ACC-M-FIX14 fixture, vacuum chuck or kinematic mount. |

## 3 Specifications · environment

| FEATURE | SPECIFICATION |
|---|---|
| Operating temperature | −50 to +200 °C. |
| Thermal expansion | Approx. 7.2 µm/(m·°C) (alumina). |
| Chemical resistance | Inert to water, alcohols, common solvents. |
| Cleaning | Lens tissue + isopropyl alcohol; no abrasives. |

## 3 Deliverables

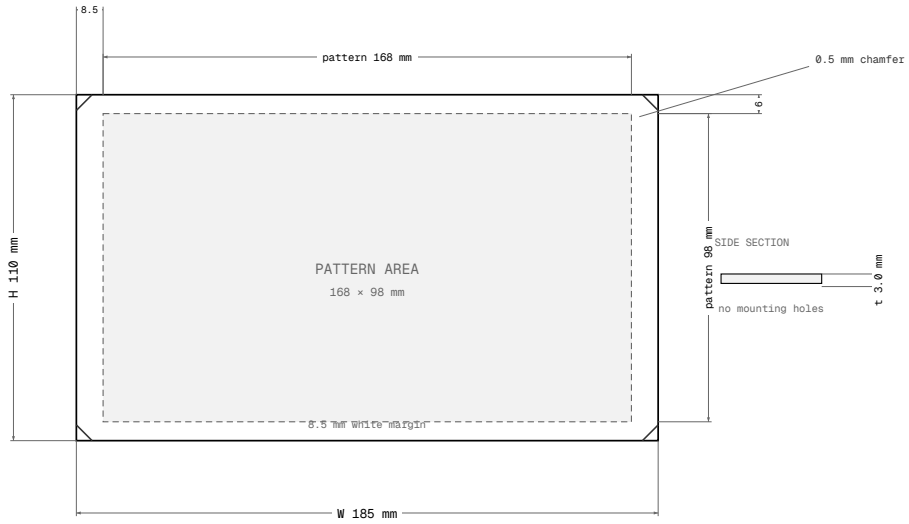
| IN THE BOX | | CONFIGURATION & SERVICE | |
|---|---|---|---|
| Board | ChArUco ceramic target, selected part number. | Marking on board | Label line with part / pattern / SN / Rev; printed X / Y arrows. |
| Packaging | Padded case and handling information. | Lead time | Made to order · ships in 2–3 weeks. |
| Measurement | Every unit is measured before it ships. | Options | Fixture · custom NRE. |
| Files | `.opencv.json` · `.ros2.yaml` · `.halcon.txt` (from the pattern generator) | | |

## 4 Compatibility

| SOFTWARE | USE / CAUTION |
|---|---|
| OpenCV | ChArUco board with `DICT_6X6_250`; this board follows the OpenCV ≥ `4.6` convention. Do not enable legacy pattern for this board. |
| Kalibr | No native ChArUco target model, so no Kalibr file is generated. Use an AprilGrid target for Kalibr. |
| HALCON | Not a HALCON plate. Pass the inner corners from `.halcon.txt` to `set_calib_data_calib_object` and the detected corners to `set_calib_data_observ_points`. |
| MATLAB | Intended for the Computer Vision Toolbox ChArUco workflow (R2024a or later—confirm release support before use). |
| ROS 2 | Use a `charuco` pattern configuration where supported by the selected `camera_calibration` package; confirm package parameters first. |

## 5 Drawing

185 × 110 mm reference variant




DXF

Available on request:
hello@calib.tools

Drawing reference only; request the DXF for fixture design.

CT-DWG-CHA-C-185110 · Rev A · dimensions nominal unless stated.

## 6 Compliance & customs

| TOPIC | STATEMENT |
|---|---|
| RoHS | RoHS-compliant materials. |
| HS code | 9031 80 |
| Packaging | Padded case inside a carton; FRAGILE. |
| Handling | FRAGILE · handle by edges; protect the patterned surface from impact and abrasion. |

## 7 Revision history

| REV | DATE | CHANGE |
|---|---|---|
| A | 2026-10 | First issue, planning values |

## 8 Disclaimer

Values are nominal unless stated as measured.